

MATERIAL SAFETY DATA SHEET

MSDS No.: Not applicable

1. Identification of the substance or mixture and of the supplier

- A. Product name:** Lacto 10 Complex
B. Use: Cosmetic ingredient
C. Company: The Garden of Naturalsolution Co., Ltd.
 12-24, Gajangsaneopseo-ro, Osan-si, Gyeonggi-do, 18103, Republic of Korea
D. Telephone: +82-31-374-5240
E. Fax: +82-31-374-5246

2. Hazards identification

A. GHS classification of the substance/mixture

Not classified according to OSHA 29 CFR 1910.1200

B. GHS label elements, including precautionary statements

Pictogram and symbol : Not applicable

Signal word : Not applicable

Hazard statements : Not applicable

Precautionary statements

Precaution : Not applicable

Treatment : Not applicable

Storage : Room temperature

Disposal : Not applicable

C. Other hazard information not included in hazard classification (NFPA)

Health : 0

Flammability : Not available

Reactivity : Not available

3. Composition/information on ingredients

INCI Name	CAS No.	EC No.	Content (%)
Water	7732-18-5	231-791-2	To 100
Butylene Glycol	107-88-0	203-529-7	40±4
Lactobacillus/Portulaca Oleracea Ferment Extract	-	-	4.5±0.45
Lactobacillus/Licorice Root Extract Ferment Filtrate	-	-	2±0.2

Lactobacillus/Lycium Chinense Fruit Extract Ferment Filtrate	-	-	1±0.1
Lactobacillus/Houttuynia Cordata Leaf Extract Ferment Filtrate	-	-	1±0.1
Lactobacillus/Panax Ginseng Root Extract Ferment Filtrate	-	-	1±0.1
Lactobacillus/Astragalus Membranaceus Root Extract Ferment Filtrate	-	-	1±0.1
Lactobacillus/Scutellaria Baicalensis Root Extract Ferment Filtrate	-	-	1±0.1
Lactobacillus/Pueraria Lobata Root Extract Ferment Filtrate	-	-	1±0.1
Lactobacillus/Angelica Keiskei Leaf/Stem Extract Ferment Filtrate	-	-	1±0.1
Saccharomyces/Camellia Sinensis Extract Ferment Filtrate	-	-	1±0.1

4. First aid measures

A. Eye contact

- In case of contact with substance, immediately flush eyes with running water at least 20 minutes.

B. Skin contact

- In case of contact with substance, immediately flush skin with running water at least 20 minutes.
- Remove and isolate contaminated clothing and shoes.
- Wash contaminated clothing and shoes before reuse.
- Get immediate medical advice/attention.

C. Inhalation

- Specific medical treatment is urgent.
- Move victim to fresh air.
- Give artificial respiration if victim is not breathing.
- Administer oxygen if breathing is difficult.

D. Ingestion

- Do not let him/her eat anything, if unconscious.
- Get immediate medical advice/attention.

E. Indication of immediate medical attention and notes for physician

- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.

F. Most important symptoms and effects, both acute and delayed

- Not available

5. Fire fighting measures

A. Suitable (and unsuitable) extinguishing media

- Suitable extinguishing media: Dry sand, dry chemical, alcohol-resistant foam, water spray, regular foam, CO₂

- Unsuitable extinguishing media: High pressure water streams

B. Specific hazards arising from the chemical

- May be ignited by heat, sparks or flames.
- Containers may explode when heated.
- If inhaled, may be harmful.
- Fire will produce irritating and/or toxic gases.

C. Special protective equipment and precautions for fire-fighters

- Dike fire-control water for later disposal; do not scatter the material.
- Move containers from fire area if you can do it without risk.
- Fire involving Tanks; Cool containers with flooding quantities of water until well after fire is out.
- Fire involving Tanks; Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
- Fire involving Tanks; Always stay away from tanks engulfed in fire.

6. Accidental release measures

A. Personal precautions, protective equipment and emergency procedures

- Eliminate all ignition sources.
- Stop leak if you can do it without risk.
- Please note that materials and conditions to avoid.
- Ventilate the area.
- Do not touch or walk through spilled material.
- Prevent dust cloud.

B. Environmental precautions and protective procedures

- Prevent entry into waterways, sewers, basements or confined areas.

C. The methods of purification and removal

- Small Spill; Flush area with flooding quantities of water. And take up with sand or other non-combustible absorbent material and place into containers for later disposal.
- Large Spill; Dike far ahead of liquid spill for later disposal.
- With clean shovel place material into clean, dry container and cover loosely; move containers from spill area.

7. Handling and storage

A. Precautions for safe handling

- Please note that materials and conditions to avoid.
- Wash thoroughly after handling.

- Please work with reference to engineering controls and personal protective equipment.
- Be careful to high temperature.

B. Conditions for safe storage

- Store in a closed container.
- Store in cool and dry place.

8. Exposure controls/personal protection

A. Occupational Exposure limits

Korea regulation : Not available

ACGIH regulation : Not available

Biological exposure index : Not available

OSHA regulation : Not available

NIOSH regulation : Not available

EU regulation : Not available

Other : Not available

B. Appropriate engineering controls

- Provide local exhaust ventilation system or other engineering controls to keep the airborne concentrations of vapors below their respective threshold limit value.

C. Personal protective equipment

Respiratory protection

- Wear NIOSH or European Standard EN 149 approved full or half face piece (with goggles) respiratory protective equipment when necessary.
- In case exposed to gaseous/liquid material, the respiratory protective equipments as follow are recommended. escape full facepiece gas mask (of use for acid gas, in case of acid gas for organic compounds) or escape half facepiece gas mask (of use for acid gas, in case of acid gas for organic compounds) or direct full facepiece gas mask (of use for acid gas, in case of acid gas for organic compounds) half facepiece gas mask (of use for acid gas, in case of acid gas for organic compounds) or powered air-purifying gas mask.
- In lack of oxygen (< 19.5%), wear the supplied-air respirator or self-contained breathing apparatus.oxygen

Eye protection

- Wear facepiece with goggles to protect.
- An eye wash unit and safety shower station should be available nearby work place.

Hand protection

- Wear chemical resistant gloves.

Body protection

- Wear appropriate protective chemical resistant clothing.

9. Physical and chemical properties

1. Appearance	Liquid
Description	
Color	Colorless ~ light yellow
2. Odor	Characteristic
3. Odor threshold	No data
4. pH(1→10)	4.0 ~ 7.0
5. Melting point / Freezing point	No data
6. Initial boiling point and boiling point range	No data
7. Flash point	> 115°C
8. Evaporation rate	No data
9. Flammability (solid, gas)	No data
10. Upper/lower flammability or explosive limits	No data
11. Vapor pressure	No data
12. Solubility (ies)	No data
13. Vapor density	No data
14. Specific gravity	0.98 ~ 1.04
15. Partition coefficient: n-octanol/water	No data
16. Self-ignition temperature	Do not ignite automatically when contacted with air in room temperature.
17. Decomposition temperature	No data
18. Viscosity	No data
19. Molecular Weight	No data

10. Stability and reactivity

A. Chemical stability and Possibility of hazardous reactions:

- Fire may produce irritating and/or toxic gases.
- If inhaled, may be harmful.

B. Conditions to avoid:

- Ignition sources (heat, sparks or flames)

C. Incompatible materials:

- Combustibles

D. Hazardous decomposition products:

- Irritating and/or toxic gases

11. Toxicological information

A. Information of Health Hazardous

Acute toxicity

Oral : Not classified (ATEmix = 57,000 mg/kg bw)

- **Water** : Rat LD₅₀ > 90,000 mg/kg

- **Butylene Glycol** : Rat LD₅₀ = 22,800 mg/kg (male)

Dermal : Not available

Inhalation : Not available

- **Butylene Glycol** : Rat (male) (No deaths at about 292 mg/m³ 8h of saturated vapour)

Skin corrosion/ irritation : Not classified

- **Butylene Glycol** : As a result of skin irritation test using human, slight skin irritation was confirmed in 2 out of 200 subjects, but no signs of stimulation were observed after the application site change.

Serious eye damage/ irritation : Not classified

- **Butylene Glycol** : As a result of eye irritation tests using rabbits, the undiluted test substance causes slight irritation to the eyes of rabbits. (OECD Guideline 405, GLP)

Respiratory sensitization : Not available

Skin sensitization : Not classified

- **Butylene Glycol** : As a result of human skin sensitization test, slight skin irritation was confirmed in 2 out of 200 subjects but no sensitivity was observed.

Carcinogenicity : Not classified

OSHA , IARC , NTP , ACGIH , KOREA-ISHL , EU CLP , KOREA-NIER, NITE: Not listed

- **Water** :

OSHA , IARC , NTP , ACGIH , KOREA-ISHL , EU CLP , KOREA-NIER, NITE: Not listed

- **Butylene Glycol** :

OSHA , IARC , NTP , ACGIH , KOREA-ISHL , EU CLP , KOREA-NIER, NITE: Not listed

In the study of carcinogenicity in rats(male/female), treatment-related side effects (carcinogenic and non-cancerous) were not observed in rats fed up to 10% of 1,3-butylene glycol by diet. (NOAEL = 5000mg/kg bw/day)

Mutagenicity : Not classified

- **Butylene Glycol** : In vivo cytogenetic / chromosomal aberration test using rats(male/female) did not cause chromosomal anomalies. (genotoxicity test in vivo after subchronic oral exposure over 3 generations)

Reproductive toxicity : Not classified

- **Water** : Reproductive toxicity test using rats shows no adverse reproductive effects.

- **Butylene Glycol** : Reproductive toxicity test using rats(male/female) does not affect fertility. The developmental toxicity test showed no teratogenic effects at 24% (12000 mg/kg bw/day) and a fetal toxic effect at 10% (5000 mg/kg bw/day).

Specific target organ toxicity (single exposure) : Not classified

- **Butylene Glycol** : Acute oral, inhalation toxicity test results in rats (male), no symptoms related to toxicity were observed.

Specific target organ toxicity (repeat exposure) : Not classified

- **Butylene Glycol** : Oral repeated dose toxicity studies with rats (male/female) showed no toxicity related symptoms. (NOAEL = 5000mg/kg bw/day)

Aspiration Hazard : Not available

12. Ecological information

A. Ecological toxicity

- Acute toxicity : Not classified

Fish

- **Butylene Glycol** : 96hr-LC₅₀ (Oryzias latipes) > 100 mg/L (Read-across; structural analogue or surrogate) (OECD Guideline 203, GLP)

crustacean

- **Butylene Glycol** : 48hr-LC₅₀ (Daphnia magna) > 1000 mg/L (Read-across; structural analogue or surrogate) (OECD Guideline 202, GLP), 21d-NOEC (Daphnia magna) > 85 mg/L (Read-across; structural analogue or surrogate) (OECD Guideline 202, GLP)

Algae

- **Butylene Glycol** : 72hr-EC₅₀ (Scenedesmus subspicatus) > 1070 mg/L (OECD Guideline 201, GLP), 72h-NOEC (Scenedesmus subspicatus) >= 1070 mg/L (OECD Guideline 201, GLP)

- Chronic toxicity : Not classified

Fish : Not available

crustacean : Not available

Algae : Not available

B. Persistence and degradability

Persistence

- **Water** : Low persistency (log Kow is less than 4 estimated.) (Log Kow = -1.38) (estimated)

- **Butylene Glycol** : Low persistency (log Kow is less than 4 estimated.) (Log Kow = -0.9) (25 °C, pH = 7.5)

Degradability : Not available

C. Bioaccumulative potential

Bioaccumulation

- **Water** : Bioaccumulation is expected to be low according to the BCF < 500 (BCF = 50) (estimated) (SRC)

- **Butylene Glycol** : Bioaccumulation is expected to be low according to the BCF < 500 (BCF = 3.2)

Biodegradation

- **Water** : Readily biodegradable
- **Butylene Glycol** : As well-biodegraded, it is expected to have low accumulation potential in living organisms (= 81% biodegradation was observed after 29 days)

D. Mobility in soil

- **Water** : Low potency of mobility to soil. (Koc = 0.06337) (estimated)
- **Butylene Glycol** : Low potency of mobility to soil. (Koc = 1)

E. Other hazardous effect : Not available

F. Hazardous to the ozone layer : Not available

13. Disposal considerations

A. Disposal method

Waste must be disposed of in accordance with federal, state and local environmental control regulations.

B. Disposal precaution

Consider the required attentions in accordance with waste treatment management regulation.

14. Transport information

A. UN Number : Not applicable

B. UN Proper shipping name : Not applicable

C. Transport Hazard class : Not applicable

D. Packing group : Not applicable

E. Marine pollutant : Not applicable

F. Special precautions

in case of fire : Not applicable

in case of leakage : Not applicable

15. Regulatory information

① KOREA Regulatory information

A. Occupational Safety and Health Act : Not regulated

B. Chemical Control Act

Water : Existing Chemical Substance

Butylene Glycol : Existing Chemical Substance

C. Safety Control of Dangerous Substances Act

Butylene Glycol : 4000ℓ Petroleum class 4-3 (water soluble liquid)

D. Wastes Control Act : Not regulated

E. Other regulation (internal and external)

Internal information

Persistent Organic Pollutants Acts : Not regulated

② Foreign Regulatory Information

External information

EU classification(classification) : Not regulated

EU classification(risk phrases) : Not regulated

EU classification(safety phrases) : Not regulated

EU SVHC list : Not regulated

EU Authorisation List : Not regulated

EU Restriction list : Not regulated

U.S.A management information (OSHA Regulation) : Not regulated

U.S.A management information (CERCLA Regulation) : Not regulated

U.S.A management information (EPCRA 302 Regulation) : Not regulated

U.S.A management information (EPCRA 304 Regulation) : Not regulated

U.S.A management information (EPCRA 313 Regulation) : Not regulated

Substance of Roterdame Protocol : Not regulated

Substance of Stockholme Protocol : Not regulated

Substance of Montreal Protocol : Not regulated

16. Other information

A. Information source and references

American Conference of Governmental Industrial Hygienists TLVs and BEIs.

EPISUITE v4.11; <http://www.epa.gov/opt/exposure/pubs/episuitedi.htm>

EU CLP; <https://echa.europa.eu/information-on-chemicals/cl-inventory-database>

IARC Monographs on the Evaluation of Carcinogenic Risks to Humans;

<http://monographs.iarc.fr>

Korea Occupational Health & Safety Agency; <http://www.kosha.net>

LookChem; <http://www.lookchem.com/>

Ministry of Public Safety and Security-Korea dangerous material inventory management system; <http://hazmat.mpss.kfi.or.kr/index.do>

National Chemicals Information System; <http://ncis.nier.go.kr/ncis/>

National Toxicology Program; <http://ntp.niehs.nih.gov/results/dbsearch/>

NIOSH Pocket Guide; <http://www.cdc.gov/niosh/npg/npgdcas.html>

REACH information on registered substances; <https://echa.europa.eu/information-on-chemicals/registered-substances>

TOMES-LOLI® <http://www.rightanswerknowledge.com/loginRA.asp>

TOMES-LOLI®; <http://www.rightanswerknowledge.com/loginRA.asp>

U.S. National library of Medicine(NLM) ChemIDplus; <http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?CHEM>

U.S. National library of Medicine(NLM) Hazardous Substances Data Bank(HSDB); <http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?HSDB>

Waste Control Act enforcement regulation attached [1]

UN Recommendations on the transport of dangerous goods 17th

Emergency Response Guidebook 2008;

http://phmsa.dot.gov/staticfiles/PHMSA/DownloadableFiles/Files/erg2008_eng.pdf

Korea Occupational Health & Safety Agency; <http://www.kosha.or.kr>

National Chemicals Information System; <https://kreach.me.go.kr/repwrt/index.do>

American Conference of Governmental Industrial Hygienists TLVs and BEIs.

EPISUITE v4.11; <http://www.epa.gov/opt/exposure/pubs/episuitedi.htm>

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National Chemicals Information System; <http://ncis.nier.go.kr/ncis/>

National Toxicology Program; <http://ntp.niehs.nih.gov/results/dbsearch/>

NIOSH Pocket Guide; <http://www.cdc.gov/niosh/npg/npgdcas.html>

REACH information on registered substances; <https://echa.europa.eu/information-on-chemicals/registered-substances>

TOMES-LOLI® <http://www.rightanswerknowledge.com/loginRA.asp>

TOMES-LOLI®; <http://www.rightanswerknowledge.com/loginRA.asp>

U.S. National library of Medicine(NLM) ChemIDplus; <http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?CHEM>

U.S. National library of Medicine(NLM) Hazardous Substances Data Bank(HSDB); <http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?HSDB>

Waste Control Act enforcement regulation attached [1]

B. Issuing date : 2024.01.08

C. Revision number and date

revision number : 0

date of the latest revision : -

D. Others

- The content is based on the latest information and knowledge that we currently possess.
 - This SDS was authored to aid buyer, processor or any other third person who handles the chemical of subject in the SDS; additionally, it does not warrant suitability of the chemical for special purposes or the commercial use of statements that approves the use of it in combination with other chemicals as well as technical or legal liabilities.
 - The content of the SDS may vary depending on the country or the region and may not coincide with the actual regulations. Therefore, the buyer or the processor of the chemical is responsible for observing responsible government's or the region's regulations.
- For more detailed information, please contact The Garden of Naturalsolution Co., Ltd.
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